



KILLARA HIGH SCHOOL

EXTENSION 2 MATHEMATICS

**YEAR 12
MARCH 2006**

Time Allowed: 3 hours

Instructions:

- * You may attempt ALL questions
- * All necessary working should be shown
- * Start each question on a new page
- * Approved calculators may be used
- * All questions are of equal value
- * Full marks may not be awarded for careless or badly arranged work
- * Standard Integrals are printed at the end of the paper

Question 1.

- (a) Let $z = a + ib$ where a and b are real numbers and $a \neq 0$.
- (i) Express $|z|$ and $\tan(\arg z)$ in terms of a and b . (2 marks)
- (ii) Express $\frac{z}{3+4i}$ in the form $x + iy$ where x and y are real. (2 marks)
- (b) (i) If $w = \frac{1+i\sqrt{3}}{2}$ show that $w^3 = -1$ (2 marks)
- (ii) Hence calculate w^{16} (1 mark)
- (c) If $z = 5 - 5i$ write z , z^2 and $\frac{1}{z^2}$ in modulus argument form. (3 marks)
- (d) Let u and v be two complex numbers where $u = -2 + i$ and v is defined by $|v| = 3$ and $\arg v = \frac{\pi}{3}$

On an Argand diagram:

- (i) Plot the points A and B representing the complex numbers u and v respectively. (2 marks)
- (ii) Plot the points C and D represented by the complex numbers $u - v$ and iu respectively. Indicate any geometric relationships between the four points A, B, C and D. (3 marks)

Question 2 over page ...

Question 2.

- (a) Find real values of a , b and c to make the following true. (3 marks)

$$\frac{2x^2 - x + 1}{(x-1)(x^2+1)} = \frac{a}{x-1} + \frac{bx+c}{x^2+1}$$

- (b) Let $x = \alpha$ be a root of the quartic polynomial $P(x) = x^4 + Ax^3 + Bx^2 + Ax + 1$, where A and B are real but α may be complex.

- (i) Show that $\alpha \neq 0$ (1 mark)

- (ii) Show that $x = \alpha$ is also a root of $Q(x) = x^2 + \frac{1}{x^2} + A(x + \frac{1}{x}) + B$. (2 marks)

- (iii) Let $u = x + \frac{1}{x}$. Show that $Q(x)$ becomes $R(u) = u^2 + Au + (B-2)$. (2 marks)

- (c) Consider the polynomial equation $x^4 + ax^3 + bx^2 + cx + d = 0$ where a , b , c and d are all integers. Suppose the equation has a root of the form ki , where k is real and $k \neq 0$.

- (i) State why the conjugate $-ki$ is also a root. (1 mark)

- (ii) Show that $c = k^2 a$ (2 marks)

- (iii) Show that $c^2 + a^2 d = abc$ (2 marks)

- (iv) If 2 is also a root of the equation, and $b = 0$, show that c is even. (2 marks)

Continued over page ...

Question 3.

- (a) (i) Write down the relations which hold between the roots α, β, γ of the equation $ax^3+bx^2+cx+d=0$, where $a \neq 0$ and the co-efficients a, b, c, d are real. (3 marks)
- (ii) Consider the equation $36x^3-12x^2-11x+2=0$. You are given that the roots α, β, γ of this equation satisfy $\alpha=\beta+\gamma$. Use Part (i) to find α . (3 marks)
- (iii) Suppose the equation $x^3+px^2+qx+r=0$ has roots λ, μ, τ which satisfy $\lambda=\mu+\tau$. Show that $p^3-4pq+8r=0$. (3 marks)
- (b) None of the roots α, β, γ of the equation $x^3+3mx+n=0$ is zero.
- (i) Obtain the monic equation whose roots are $\frac{\beta\gamma}{\alpha}, \frac{\alpha\gamma}{\beta}$ and $\frac{\alpha\beta}{\gamma}$, expressing its co-efficients in terms of m and n . (3 marks)
- (ii) Hence show that $\gamma=\alpha\beta$ if and only if $(3m-n)^2+n=0$. (3 marks)

Question 4.

The hyperbola H has equation $xy = 16$.

- (a) Sketch this hyperbola and indicate on your diagram the positions and coordinates of all points at which the curve intersects the axes of symmetry. (3 marks)
- (b) $P(4p, \frac{4}{p})$, where $p > 0$ and $Q(4q, \frac{4}{q})$ where $q > 0$ are two distinct arbitrary points on H . Find the equation of the chord PQ . (2 marks)
- (c) Prove that the equation of the tangent at P is $x+p^2y=8p$. (2 marks)
- (d) The tangents at P and Q intersect at T . Find the coordinates of T . (3 marks)
- (e) The chord PQ produced passes through the point $N(0, 8)$.
- (i) Find the equation of the locus of T . (3 marks)
- (ii) Give the geometrical description of this locus. (2 marks)

Continued over page ...

Question 5.

- (a) (i) Suppose that k is a triple root of the polynomial $P(x)=0$. Show that $P''(k)=0$. (2 marks)
- (ii) What feature does the graph of a polynomial have at a root of multiplicity 3? (1 mark)
- (iii) The polynomial $Q(x)=x^4+4x^3-16x-16$ has a root of multiplicity 3. Solve $Q(x)=0$. (2 marks)
- (iv) Let $R(x)=1+\frac{x}{1}+\frac{x^2}{1\times 2}+\frac{x^3}{1\times 2\times 3}+\frac{x^4}{1\times 2\times 3\times 4}+\frac{x^5}{1\times 2\times 3\times 4\times 5}$.

Prove that $R(x)$ has no triple root. (2 marks)

- (b) The cubic equation $x^3+px+q=0$ has 3 non zero roots α, β, γ . Find in terms of the constants p and q the values of
- (i) $\alpha^2+\beta^2+\gamma^2$ (2 marks)
- (ii) $\alpha^2\beta^2+\beta^2\gamma^2+\alpha^2\gamma^2$ (2 marks)
- (iii) Show that the condition for the roots α, β, γ to be in geometric progression is that $p=0$. (2 marks)
- (iv) Prove that these roots cannot be in Arithmetic Progression. (2 marks)

Continued over page ...

Question 6.

(a) Evaluate i^{2006} (1 mark)

(b) Let $z = \frac{18+4i}{3-i}$

(i) Simplify $(18+4i)(3-i)$ (1 mark)

(ii) Express z in the form $a+ib$ where a and b are real numbers. (2 marks)

(iii) Hence or otherwise find $|z|$ and $\arg(z)$. (2 marks)

(c) Sketch the region where the inequalities $|z-3+i| \leq 5$ and $|z+1| \leq |z-1|$ both hold. (3 marks)

(d) Let $w = \frac{3+4i}{5}$ and $z = \frac{5+12i}{13}$

(i) Show that $|w| = |z| = 1$ (2 marks)

(ii) Find wz and $w\bar{z}$ in the form $a+ib$ (2 marks)

(iii) Hence find two distinct ways of writing 65^2 as the sum p^2+q^2 where p and q are integers and $0 < p < q$. (2 marks)

Question 7.

The ellipse $E : \left(\frac{x}{5}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$ has foci $S(4,0)$ and $S'(-4,0)$

(i) Sketch the ellipse E indicating its foci S, S' and its directrices. (3 marks)

(ii) Show that the tangent at $P(x_1, y_1)$ on the ellipse E has equation $9x_1x + 25y_1y = 225$. (3 marks)

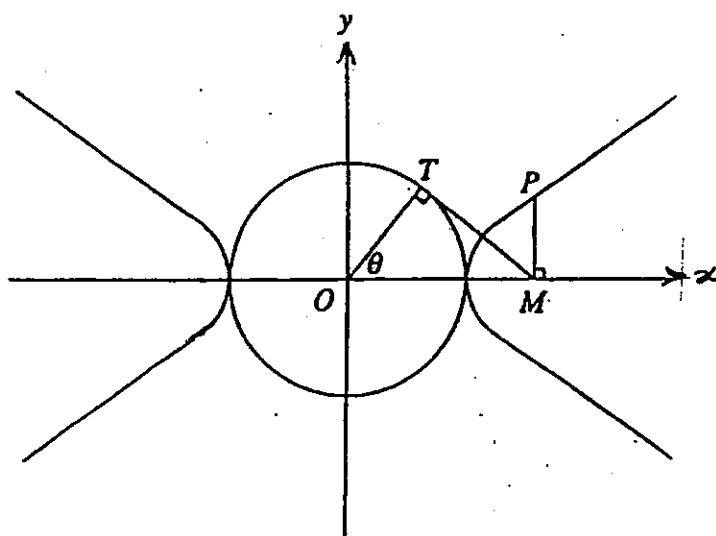
(iii) The line joining $P(x_1, y_1)$ to $Q(x_2, y_2)$ passes through S . Show that $4(y_2 - y_1) = x_1y_2 - x_2y_1$. (3 marks)

(iv) It is also known that $Q(x_2, y_2)$ lies on E . Show that the tangents at P and Q on the ellipse intersect on the directrix corresponding to S . (3 marks)

(v) Find the equation of the normal to E at P and decide under what circumstances, if any, it passes through S or S' . (3 marks)

Question 8.

(a)

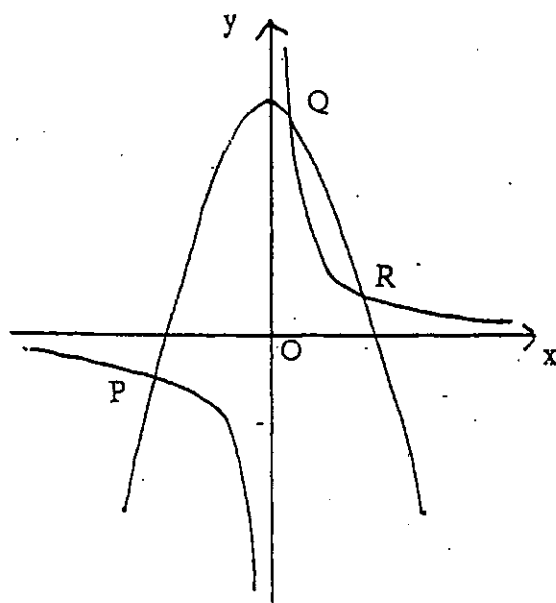


The figure shows the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and the circle $x^2 + y^2 = a^2$, where $a, b > 0$. The point T lies on the circle with $\angle TOx = \theta$, where $0 < \theta < \frac{\pi}{2}$. The tangent to the circle at T meets the x axis at M ; MP is perpendicular to Ox and P is a point on the hyperbola in the first quadrant.

- (i) Show that P has coordinates $(a \sec \theta, b \tan \theta)$. (2 marks)
- (ii) Suppose that Q is a point on the hyperbola with coordinates $(a \sec \phi, b \tan \phi)$. If $\theta + \phi = \frac{\pi}{2}$, and $\theta \neq \frac{\pi}{4}$, show that the chord PQ has equation $ay = b(\cos \theta + \sin \theta)x - ab$. (3 marks)
- (iii) Show that every such chord PQ passes through a fixed point and find its coordinates. (2 marks)
- (iv) Show that as θ approaches $\frac{\pi}{2}$ the chord PQ approaches a line parallel to an asymptote. (2 marks)

Question 8(b) over page...

(b)



The curves $y = k - x^2$, for some real number k , and $y = \frac{1}{x}$ intersect at the points P, Q and R where $x = \alpha$, $x = \beta$ and $x = \gamma$.

- (i) Show that the monic cubic equation with coefficients in terms of k whose roots are α^2 , β^2 and γ^2 is given by $x^3 - 2kx^2 + k^2x - 1 = 0$. (2 marks)
- (ii) Find the monic cubic equation with coefficients in terms of k whose roots are $\frac{1}{\alpha^2}$, $\frac{1}{\beta^2}$ and $\frac{1}{\gamma^2}$ (2 marks)
- (iii) Hence show that $OP^2 + OQ^2 + OR^2 = k^2 + 2k$, where O is the origin. (2 marks)

QUESTION 1.

(a) (i) $z = a + ib$

$|z| = \sqrt{a^2 + b^2}$, $\tan(\arg z) = \frac{b}{a}$

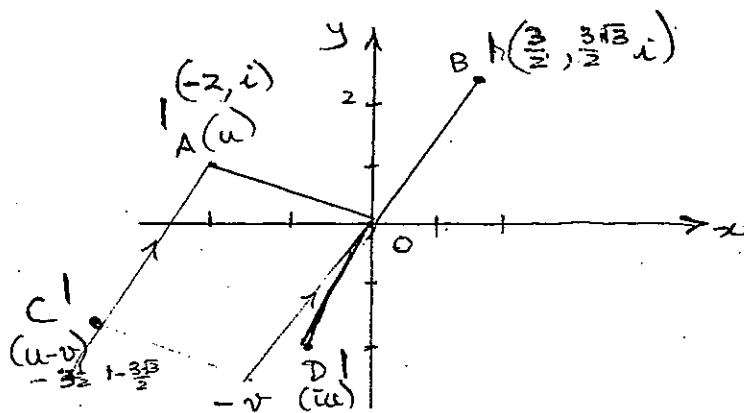
(ii) $\frac{z}{3+4i} = \frac{(a+ib)(3-4i)}{3^2+4^2}$
 $= \frac{3a+4b}{25} + i \frac{(-4a+3b)}{25}$

(b) (i) $w = \frac{1+i\sqrt{3}}{2} = 1(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$
 $w^3 = 1(\cos \pi + i \sin \pi)$
 $= -1$

(ii) $w^{16} = w^{15} w$
 $= (-1)^5 \frac{1+i\sqrt{3}}{2}$
 $= -\frac{1}{2} - i \frac{\sqrt{3}}{2}$
 Accept $- \cos \frac{\pi}{3}$

(c) $z = 5 - 5i$
 $= 5\sqrt{2} \left(\cos \left(-\frac{\pi}{4} \right) + i \sin \left(-\frac{\pi}{4} \right) \right)$
 $z^2 = 50 \left[\cos \left(-\frac{\pi}{2} \right) + i \sin \left(-\frac{\pi}{2} \right) \right]$
 $\frac{1}{z^2} = (z^2)^{-1}$
 $= \frac{1}{50} \left[\cos \left(\frac{\pi}{2} \right) + i \sin \left(\frac{\pi}{2} \right) \right]$
 $= \frac{i}{50}$

(d) $u = -2 + i$ $v = 3(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$
 $= 3 \left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right)$



ABOC is a parallelogram,
 $OD \perp OA$.

QUESTION 2.

(a) $\frac{2x^2 - x + 1}{(x-1)(x^2+1)} = \frac{a}{x-1} + \frac{bx+c}{x^2+1}$

$\therefore 2x^2 - x + 1 = a(x^2+1) + (bx+c)(x-1)$
 $= a(x^2+1) + bx^2 - bx + cx - c$
 $= x^2(a+b) + x(-b+c) + a - c$

$\therefore a+b = 2$ — (1)

$-b+c = -1$ — (2)

$a-c = 1$ — (3)

(1)+(2) $a+c = 1$ — (4)

(3)+(4) $2a = 2 \therefore a = 1$

Sub in (1) $1+b = 2 \therefore b = 1$

Sub in (3) $1-c = 1 \therefore c = 0$

$a = 1, b = 1, c = 0$

(b) $P(x) = x^4 + Ax^3 + Bx^2 + Ax + 1$

$P(x) = x^4 + Ax^3 + Bx^2 + Ax + 1$

true if x is a root $P(x) = 0$

but $P(0) = 1 \therefore x = 0$ is not a root

QUEST 2 (b) (ii)

$$Q(x) = x^2 + \frac{1}{x^2} + A(x + \frac{1}{x}) + B$$

$$1 = \frac{1}{x^2} [x^4 + 1 + Ax^2(x + \frac{1}{x}) + Bx^2]$$

$$= \frac{1}{x^2} [x^4 + 1 + Ax^3 + Ax + Bx^2]$$

$$= \frac{P(x)}{x^2}$$

$$\therefore Q(x) = \frac{P(x)}{x^2} \quad \text{Note } x \neq 0$$

$$\therefore x = \alpha \text{ is a root of } Q(x)$$

(iii) Let $u = x + \frac{1}{x}$

$$(x + \frac{1}{x})^2 = (x^2 + \frac{1}{x^2}) + 2$$

$$\therefore Q(x) = (x + \frac{1}{x})^2 + A(x + \frac{1}{x}) + B - 2$$

$$= u^2 + A(u) + B - 2$$

$$= R(u).$$

(c) $x^4 + ax^3 + bx^2 + cx + d = 0$

(i) For polynomials with real co-efficients, complex roots occur in conjugate pairs. I.e. $(a - ib)$ a factor then $(a + ib)$ is a factor $\Rightarrow a^2 + b^2$ a real factor

(ii) $P(x) = x^4 + ax^3 + bx^2 + cx + d$

$$P(ki) = k^4 - ak^3i - bk^2 +cki + d$$

if factor $\neq 0 = P(ki)$

$$\therefore \text{Imag } 0 = -ak^3 + ck$$

$$\therefore c = k^2a$$

(iii) Equating Real Parts

$$k^4 - bk^2 + d = 0$$

But $k^2 = \frac{c}{a}$

$$\therefore (\frac{c}{a})^2 - b(\frac{c}{a}) + d = 0$$

$$c^2 - abc + a^2d = 0$$

$$c^2 + a^2d = abc$$

(iv)

$$\therefore 16 + 8a + 4b + 2c + d = 0$$

From $b = 0$

$$16 + 8a + 2c + d = 0$$

and $c^2 + a^2d = a \times 0 \times c$

$$\therefore d = -\frac{c^2}{a^2}$$

$$\therefore 16 + 8a + 2c - \frac{c^2}{a^2} = 0$$

I.e. $c^2 - 2a^2c - 8a^3 - 16a^2 = 0$

Solve for c

$$c = \frac{2a^2 \pm \sqrt{4a^4 + 4(8a^3 + 16a^2)}}{2}$$

$$= a^2 \pm \sqrt{a^4 + 8a^3 + 16a^2}$$

$$= a^2 \pm a\sqrt{(a+4)^2}$$

I.e. $C = (a^2 + a^2 + 4a)$ or $a^2 - a^2 - 4a$

$$= 2(a^2 + 4a) \text{ or } 2(-2a)$$

I.e. $\therefore$ is even. (a is an integer)

Q8 $P(x)$ is degree 4 with roots $ki, -ki, 2$ and α .

$$\therefore \sum \alpha = 2 + \alpha = -a$$

$$\sum \alpha \beta = k^2 + 2\alpha = 0$$

$$\therefore k^2 = -2\alpha$$

$$= 2(a+2)$$

But $k^2 = \frac{c}{a} \therefore \frac{c}{a} = 2(a+2)$

$$c = 2(a^2 + 2a)$$

$$= \text{even.}$$

QUESTION 3.

a) (i) $\Sigma x = x + \beta + \gamma = -b/a$ |
 $\Sigma x\beta = x\beta + \beta\gamma + \gamma x = c/a$ |
 $x\beta\gamma = -d/a$ |

(ii) $36x^3 - 12x^2 - 11x + 2 = 0$
 Has roots $\beta, \gamma, (\beta + \gamma)$
 $\therefore 12\beta + 2\gamma = \frac{12}{36} = \frac{1}{3}$ ①
 $x = \beta + \gamma = \frac{1}{6}$ |

(iii) $x^3 + px^2 + qx + r = 0$
 Roots: λ, μ, τ
 $\lambda = \mu + \tau$
 $\Sigma x: 2\mu + 2\tau = -p$
 $\lambda = \mu + \tau = -p/2$ |

$\Sigma x\beta: (\mu + \tau)\mu + \mu\tau + (\mu + \tau)\tau = q$
 $(\mu + \tau)(\mu + \tau) + \mu\tau = q$
 $\frac{p^2}{4} + \mu\tau = q$
 NOT NEEDED

But if λ is a root:

$\lambda^3 + p\lambda^2 + q\lambda + r = 0$
 $(-p/2)^3 + p(-p/2)^2 + q(-p/2) + r = 0$
 $-p^3/8 + p^3/4 - p^2q/2 + r = 0$
 $p^3 - 4p^2q + 8r = 0$ |

Results used here are also in Q5(b)

(b) $x^3 + 3mx + n = 0$

(i) $\Sigma x = 0$ $\Sigma x\beta = 3m$ $x\beta\gamma = -n$.

$\frac{x\beta}{x} + \frac{x\gamma}{\beta} + \frac{x\beta}{\gamma} = \frac{(x\beta)^2 + (x\gamma)^2 + (x\beta)^2}{x\beta\gamma}$

Sum 1 at a time = $\frac{(x\beta)^2 + (x\gamma)^2 + (x\beta)^2}{x\beta\gamma}$

Now $(x\beta + x\gamma + \beta\gamma)^2$
 $= x^2\beta^2 + x^2\gamma^2 + \beta^2\gamma^2 + 2(x\beta\gamma) + \beta(x\beta\gamma) + \gamma(x\beta\gamma)$
 $+ 2(x\beta\gamma) + 2(x\beta\gamma) + \beta(x\beta\gamma) + \gamma(x\beta\gamma)$

i.e. $x^2\beta^2 + x^2\gamma^2 + \beta^2\gamma^2$
 $= (x\beta + x\gamma + \beta\gamma)^2 - 2x\beta\gamma \Sigma x$
 $= (3m)^2 - 2 \times n \times 0 = 9m^2$

$\therefore$ Sum 1 at a time = $-\frac{9m^2}{n}$ |

and considering 2 at a time

$(\frac{x\beta}{\gamma} \times \frac{x\gamma}{\beta}) + (\frac{x\beta}{\gamma} \times \frac{\beta\gamma}{x}) + (\frac{\beta\gamma}{x} \times \frac{x\gamma}{\beta})$

$= x^2 + \beta^2 + \gamma^2$

$= (x + \beta + \gamma)^2 - 2 \Sigma x\beta$

$= 0^2 - 2 \times 3m$

$= -6m$ |

and 3 at a time

$\frac{x\beta}{\gamma} \times \frac{x\gamma}{\beta} \times \frac{\beta\gamma}{x} = x\beta\gamma$

$= -n$ |

$\therefore$ Polynomial (must be monic) is

$x^3 + \frac{9m^2}{n}x^2 - 6mx + n = 0$

(ii) If $\gamma = x\beta$ $\therefore$ the root $\frac{x\beta}{\gamma} = 1$ |

$\therefore 1 + \frac{9m^2}{n} - 6m + n = 0$ |

$n + 9m^2 - 6mn + n^2 = 0$

$(3m - n)^2 + n = 0$ |

b(i)

Method 2: $x\beta\gamma = -n$. But

But $\frac{x\beta}{\gamma} = \frac{x\beta\gamma}{\gamma^2} = \frac{-n}{\gamma^2}$

Let $Y = \frac{-n}{x^2} \therefore x^2 = \frac{-n}{Y}$

$P(x) = x^3 + 3mx + n = 0$

$x(x^2 + 3m) = -n$

$x^2(x^2 + 3m) = -n^2$

$\frac{-n}{Y}(-\frac{n}{Y} + 3m) = -n^2$

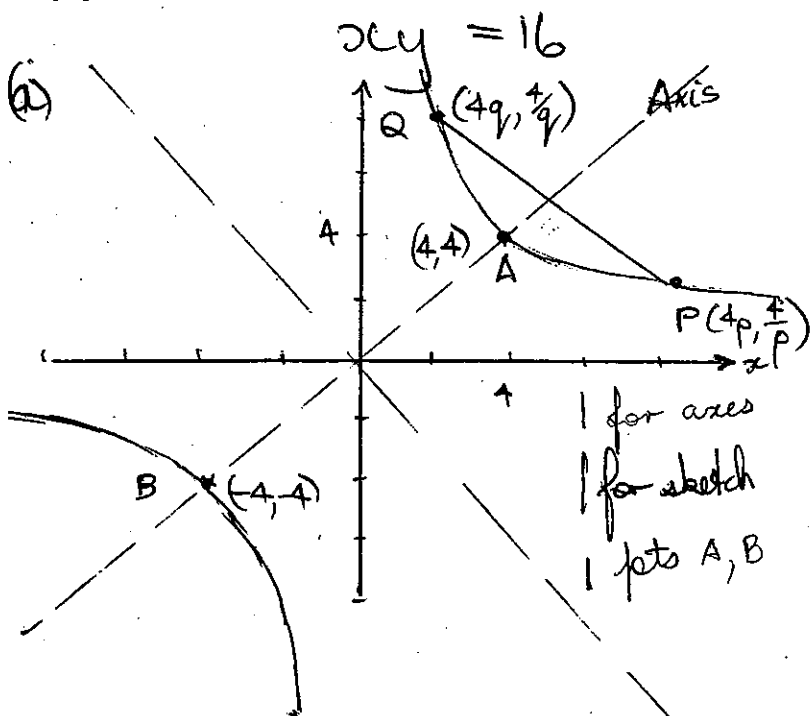
$\frac{-n}{Y}(9m^2 - \frac{6mn}{Y} + \frac{n^2}{Y^2}) = -n^2$

$-9m^2nY^2 + 6mn^2Y - n^3 = n^2Y^3$

$n^2Y^3 + 9m^2nY^2 - 6mn^2Y + n^3 = 0$

$\therefore x^3 + \frac{9m^2}{n}x^2 - 6mx + n = 0$

QUESTION 4



Intersects @ $A = (4, 4)$ $B = (-4, -4)$

(b) $P(4p, \frac{4}{p})$ $Q(4q, \frac{4}{q})$

$$m_{PQ} = \frac{\frac{4}{q} - \frac{4}{p}}{4q - 4p} \div 4 \times pq$$

$$= \frac{(p-q)}{(q-p)4pq}$$

$$= -\frac{1}{pq}$$

Chord is $y - \frac{4}{p} = -\frac{1}{pq}(x - 4p)$

$$pqy - 4q = -x + 4q$$

$$x + pqy = 4(p+q)$$

(c) Tangent $Q \rightarrow P \therefore q \rightarrow p$

$$x + p^2y = 8p$$

is equation of tangent

(d)

$$x + p^2y = 8p$$

$$x + q^2y = 8q$$

$$\therefore y(p^2 - q^2) = 8(p - q)$$

$$\therefore y = \frac{8}{p+q}$$

$$\therefore x + \frac{8p^2}{p+q} = 8p$$

$$x = \frac{8p(p+q) - 8p^2}{p+q}$$

$$= \frac{8pq}{p+q}$$

T is $(\frac{8pq}{p+q}, \frac{8}{p+q})$

(e) Through $(0, 8)$

$$\therefore 0 + 8pq = 4(p+q)$$

$$\text{or } 2pq = p+q$$

So T is $(\frac{8pq}{2pq}, \frac{8}{p+q})$

$$\therefore x = 4 \text{ is locus}$$

Note $x = pqy$ with 1

(ii) Note: Given $p > 0$ and $q > 0$

and $2pq = p+q$

$$\therefore q = \frac{p}{2p-1}$$

If $p > 1$

$$q = \frac{1+k}{2k+1} = \frac{2k+1}{2k+1} - \frac{k}{2k+1} < 1$$

i.e. $4p > 4 \therefore P$ to right of A

$4q < 4 \therefore Q$ to left of A

$\therefore T$ is between $xy=16$ and axes

$\therefore$ Locus is intercept of $x=4$ from x -axis to A

QUESTION 5

$$\begin{aligned}
 (a) (i) \quad P(x) &= (x-k)^3 \times Q(x) \\
 \therefore P'(x) &= 3(x-k)^2 Q(x) + (x-k)^3 Q'(x) \\
 &= (x-k)^2 [3Q(x) + (x-k)Q'(x)] \\
 &= (x-k)^2 \times R(x) \\
 \therefore P''(x) &= 2(x-k)R(x) + (x-k)^2 R'(x) \\
 &= (x-k)[2R(x) + (x-k)R'(x)] \\
 \therefore P''(k) &= 0
 \end{aligned}$$

(ii) If root of multiplicity 3 then $P(x)$ has a horizontal point of inflexion on the x -axis.

$$(iii) Q(x) = x^3 + 4x^2 - 16x - 16$$

$$Q'(x) = 4x^2 + 8x - 16$$

$$Q''(x) = 8x + 8$$

$$\text{If } Q''(x) = 0 = 8x + 8$$

$$x = -1 \text{ or } -2$$

$$\text{Now } Q(-1) = -1 - 4 + 16 - 16 = -5 \neq 0$$

$$Q(-2) = -8 + 16 - 32 - 16 = -40 \neq 0$$

$$\therefore Q(x) = (x+2)^3(x-\alpha)$$

$$\therefore 8(-\alpha) = -16$$

$$\alpha = 2$$

$$\therefore x = -2 \text{ or } 2$$

$$(iv) R(x) = 1 + \frac{x}{1} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^5}{5!}$$

$$\therefore R'(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}$$

$$\therefore R''(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!}$$

But $R(x)$ and $R''(x)$ have no common factor $\therefore R(x) \neq 0$ when

$$R''(x) = 0 \Rightarrow \text{No Triple root}$$

$$(b) \quad x^3 + px^2 + qx + r = 0 \text{ has 3 non-zero roots } \alpha, \beta, \gamma$$

$$\Sigma \alpha = 0 \quad \Sigma \alpha\beta = p \quad \Sigma \alpha\beta\gamma = -r$$

$$\begin{aligned}
 (i) \therefore \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2\Sigma \alpha\beta \\
 &= 0 - 2p \\
 &= -2p
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \alpha^2\beta^2 + \beta^2\gamma^2 + \alpha^2\gamma^2 & \text{ see Q3.1} \\
 \text{Consider } (\alpha\beta + \beta\gamma + \alpha\gamma)^2 & \\
 = \alpha^2\beta^2 + \alpha\beta^2\gamma + \alpha^2\beta\gamma & \\
 + \beta^2\gamma^2 + \alpha\beta^2\gamma & + \alpha\beta\gamma^2 \\
 + \alpha\gamma^2 & + \alpha^2\beta\gamma + \alpha\beta\gamma^2 \\
 = \Sigma(\alpha\beta)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma) & \\
 = p^2 - 2 \times -r \times 0 & \\
 = p^2 &
 \end{aligned}$$

$$\begin{aligned}
 (iii) \quad \text{For Geometric } \frac{\alpha}{\beta} &= \frac{\beta}{\gamma} \\
 \therefore \alpha\gamma &= \beta^2 \\
 \text{But } \alpha\gamma + \alpha\beta + \beta\gamma &= p \\
 \therefore \beta^2 + \beta(\alpha + \gamma) &= p \\
 \therefore \beta(\alpha + \beta + \gamma) &= p \\
 \therefore p &= 0
 \end{aligned}$$

$$\begin{aligned}
 (iv) \quad \text{For arithmetic } \beta - \alpha &= \gamma - \beta \\
 \therefore \alpha + \gamma - 2\beta &= 0 \\
 \therefore (\alpha + \beta + \gamma) - 3\beta &= 0 \\
 \therefore 3\beta &= 0 \\
 \text{But if } \alpha, \beta, \gamma &\text{ are non-zero} \\
 \therefore \text{cannot be arithmetic} &
 \end{aligned}$$

QUESTION 6.

$$(a) i^2 = -1 \quad i^4 = 1$$

$$\therefore i^{2006} = i^{2004} \times i^2$$

$$= -1$$

$$(b)(i) (18+4i)(3-i)$$

$$= (18+4i)(3+i)$$

$$= (54-4) + i(18+12)$$

$$= 50 + 30i$$

$$(ii) Z = \frac{18+4i}{3-i} \times \frac{3+i}{3+i}$$

$$= \frac{50+30i}{10}$$

$$= 5+3i$$

$$(iii) |Z| = \sqrt{5^2+3^2}$$

$$= \sqrt{34}$$

$$\arg Z = \tan^{-1}\left(\frac{3}{5}\right)$$

$$\approx 0.540 \text{ radians}$$

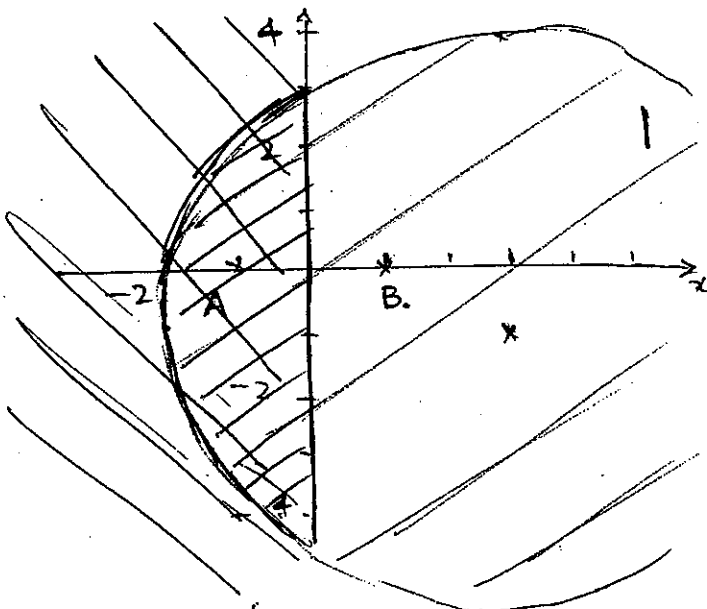
$$|Z-3+i| \leq 5 \quad \text{Centre } (3,-1)$$

$$\text{Radius } \leq 5$$

$$|Z+1| \leq |Z-1|$$

$$\text{If } A = (-1,0) \quad B = (1,0)$$

Points closer to A than B.



$$(d) w = \frac{3+4i}{5} \quad z = \frac{5+12i}{13}$$

$$(i) |w| = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2}$$

$$= \sqrt{\frac{9+16}{25}}$$

$$= 1$$

$$(ii) |z| = \sqrt{\left(\frac{5}{13}\right)^2 + \left(\frac{12}{13}\right)^2}$$

$$= \sqrt{\frac{25+144}{169}}$$

$$= 1$$

$$\text{i.e. } |w| = |z| = 1$$

$$(ii) wz = \left(\frac{3+4i}{5}\right) \times \left(\frac{5+12i}{13}\right)$$

$$= \frac{15-48+i(36+20)}{65}$$

$$= \frac{-33+56i}{65}$$

$$w\bar{z} = \left(\frac{3+4i}{5}\right) \times \left(\frac{5-12i}{13}\right)$$

$$= \frac{15+48+i(-36+20)}{65}$$

$$= \frac{63-16i}{65}$$

$$(iii) |wz| = |w||z| = 1$$

$$\therefore 33^2 + 56^2 = 65^2$$

$$\text{also } |w\bar{z}| = |w||\bar{z}| = 1$$

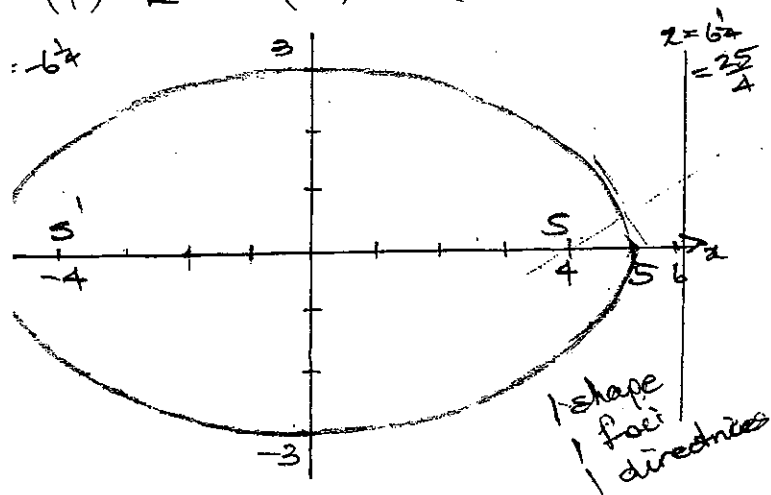
$$\therefore 63^2 + 16^2 = 65^2$$

$$\text{i.e. } p=33 \quad q=56$$

$$\text{or } p=16 \quad q=63$$

QUESTION 7

$$(i) E: \left(\frac{x}{5}\right)^2 + \left(\frac{y}{3}\right)^2 = 1.$$



$$a=5 \quad b=3$$

$$b^2 = a^2(1-e^2)$$

$$9 = 25(1-e^2)$$

$$e^2 = \frac{16}{25} \quad e = \frac{4}{5}$$

$$\text{Directrices are } x = \frac{25}{4} = 6\frac{1}{4}.$$

$$(ii) \frac{x^2}{25} + \frac{y^2}{9} = 1.$$

$$\frac{2x}{25} + \frac{2y}{9} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{25} \times \frac{9}{y}.$$

Tangent @ P

$$y - y_1 = \frac{-9x_1}{25y_1} (x - x_1)$$

$$25yy_1 - 25y_1^2 = -9x_1x + 9x_1^2$$

$$9xx_1 + 25yy_1 = 9x_1^2 + 25y_1^2$$

$$\text{i.e. } 9xx_1 + 25yy_1 = 225$$

$$(\text{from } E: 9x_1^2 + 25y_1^2 = 225)$$

$$(iii) P(x_1, y_1) \text{ to } Q(x_2, y_2)$$

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1} \text{ through } (4, 0)$$

$$\therefore \frac{-y_1}{4 - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Method 2: } m_{PQ} = m_{QS}$$

$$\therefore (4 - x_1)(y_2 - y_1) = -y_1(x_2 - x_1)$$

$$\text{i.e. } 4(y_2 - y_1) = x_1y_2 - x_1y_1 - y_1x_2 + x_1y_1$$

$$4(y_2 - y_1) = x_1y_2 - y_1x_2$$

$$(iv) Q(x_2, y_2) \text{ on ellipse}$$

$$\text{Tangent @ P (or use (ii) equation)}$$

$$\left. \begin{aligned} \frac{xx_1}{25} + \frac{yy_1}{9} &= 1 \quad \text{--- (1)} \\ \frac{xx_2}{25} + \frac{yy_2}{9} &= 1 \quad \text{--- (2)} \end{aligned} \right\}$$

$$\text{From (1) } \frac{yy_1}{9} = 1 - \frac{xx_1}{25}$$

$$y = \frac{9}{y_1} \left(\frac{25 - xx_1}{25} \right)$$

$$\text{Sub in (2) } \frac{xx_2}{25} + \frac{y_2}{9} \times \frac{9}{y_1} \left(\frac{25 - xx_1}{25} \right) = 1$$

$$xx_2 + \frac{y_2}{y_1} (25 - xx_1) = 25$$

$$xx_2 + \frac{y_2}{y_1} 25 - \frac{xx_1y_2}{y_1} = 25$$

$$x(x_2 - \frac{y_2x_1}{y_1}) = 25(1 - \frac{y_2}{y_1})$$

$$x = 25 \left(\frac{y_1 - y_2}{y_1} \right) \times \left(\frac{y_1}{x_2y_1 - y_2x_1} \right)$$

$$= \frac{25(y_1 - y_2) \times y_1}{-4(y_2 - y_1)} \text{ from (iii)}$$

$$x = +\frac{25}{4} \text{ i.e. Directrix!}$$

$$(v) \text{ Normal } y - y_1 = \frac{25y_1}{9x_1} (x - x_1)$$

$$\text{If } S(4, 0) - y_1, 9x_1 = 25y_1(4 - x_1)$$

$$-9x_1y_1 = 100y_1 - 25x_1y_1$$

$$16x_1y_1 = 100y_1 \quad y_1 \neq 0$$

$$\therefore x_1 = \frac{25}{4}$$

i.e. outside ellipse

$\therefore$ impossible.

$$\text{If } S'(-4, 0) \quad x_1 = \frac{100}{34} \text{ possible}$$

QUESTION 8:

(a) (i) $T = (a \cos \theta, a \sin \theta)$
as on circle $x^2 + y^2 = a^2$

The tangent TM is

$$x \cos \theta + y \sin \theta = a$$

$$\therefore x \cos \theta + y \sin \theta = a \quad \Bigg\} 1$$

At M $y = 0$

$$\therefore x \cos \theta = a$$

$$x = a \sec \theta$$

This is also x-co-ord of P

$$\therefore \frac{a^2 \sec^2 \theta}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{y^2}{b^2} = \sec^2 \theta - 1$$

$$y^2 = b^2 \tan^2 \theta$$

$$y = \pm b \tan \theta$$

$P = (a \sec \theta, b \tan \theta)$ as shown

(ii) $Q = (a \sec \phi, b \tan \phi)$

$\therefore$ Chord PQ is

$$\frac{y - b \tan \theta}{x - a \sec \theta} = \frac{b \tan \phi - b \tan \theta}{a \sec \phi - a \sec \theta}$$

But $\phi = \frac{\pi}{2} - \theta$

$$\therefore \frac{y - b \tan \theta}{x - a \sec \theta} = \frac{b \cot \theta - b \tan \theta}{a \csc \theta - a \sec \theta}$$

$$= \frac{b}{a} \times \frac{\frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta}}{\frac{1}{\sin \theta} - \frac{1}{\cos \theta}} \quad \Bigg\} 1$$

$$= \frac{b}{a} \times \frac{\cos^2 \theta - \sin^2 \theta}{\cos \theta - \sin \theta}$$

$$= \frac{b}{a} (\cos \theta + \sin \theta)$$

$$\therefore ay - ab \tan \theta = b(x - a \sec \theta)(\cos \theta + \sin \theta)$$

$$= bx(\cos \theta + \sin \theta) - ba(1 + \tan \theta)$$

$$\therefore ay = bx(\cos \theta + \sin \theta) - ba$$

i.e. $ay = b(\cos \theta + \sin \theta)x - ab$

(iii) $y = \frac{b}{a}(\cos \theta + \sin \theta)x - b$

Says for all values of θ
line has y-intcpt $-b$
i.e. passes thro' $(0, -b)$

(iv) as $\theta \rightarrow \frac{\pi}{2}$ $\phi \rightarrow 0$

Chord $\rightarrow y = \frac{b}{a}(0+1)x - b$

i.e. $y = \frac{b}{a}x - b$

which is $\parallel$ to $y = \frac{b}{a}x$
which is an asymptote.

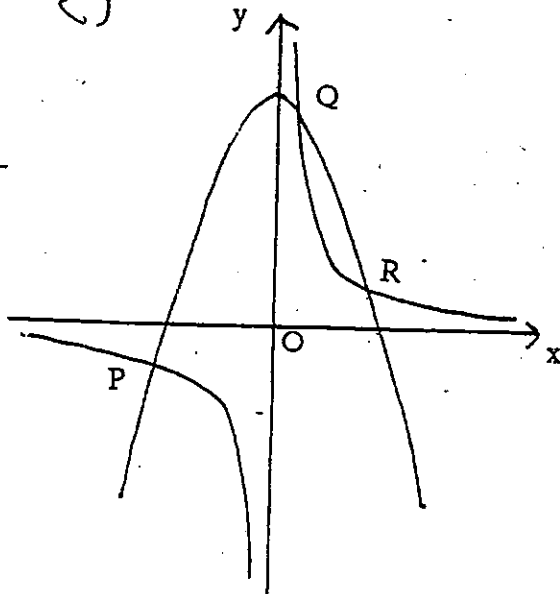
8(a)(i) In ΔOTM $\cos \theta = \frac{OT}{OM}$

$$\therefore OM = a \sec \theta = P_x$$

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QUESTION 8

(b) $y = k - x^2$ (k real)
 $y = \frac{1}{x}$



Intersect @ P, Q, R

$P = (\alpha, \frac{1}{\alpha})$ $Q = (\beta, \frac{1}{\beta})$ $R = (\gamma, \frac{1}{\gamma})$

(i) α, β, γ are roots of
 $k - x^2 = \frac{1}{x}$

i.e. $x^3 - kx + 1 = 0$

For roots $\alpha^2, \beta^2, \gamma^2$ Let $x = \sqrt{x}$

$\therefore (\sqrt{x})^3 - k(\sqrt{x}) + 1 = 0$

$\sqrt{x}(x - k) = -1$

$(x - k)^2 \cdot x = 1$

$x^3 - 2kx^2 + k^2x - 1 = 0$

(ii) The above has $\alpha^2, \beta^2, \gamma^2$

To find roots $\frac{1}{\alpha^2}, \frac{1}{\beta^2}, \frac{1}{\gamma^2}$

Let $x = \frac{1}{x}$

$\therefore (\frac{1}{x})^3 - 2k(\frac{1}{x})^2 + k^2(\frac{1}{x}) - 1 = 0$

$1 - 2kx + k^2x^2 - x^3 = 0$

i.e. $x^3 - k^2x^2 + 2kx - 1 = 0$

(iii) $OP^2 + OQ^2 + OR^2$
 $= (\alpha^2 + \frac{1}{\alpha^2}) + (\beta^2 + \frac{1}{\beta^2}) + (\gamma^2 + \frac{1}{\gamma^2})$
 $= (\alpha^2 + \beta^2 + \gamma^2) + (\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2})$
 (Sums of roots previous eqn) $\frac{2}{1}$
 $= +2k + k^2$
 $= k^2 + 2k$

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